

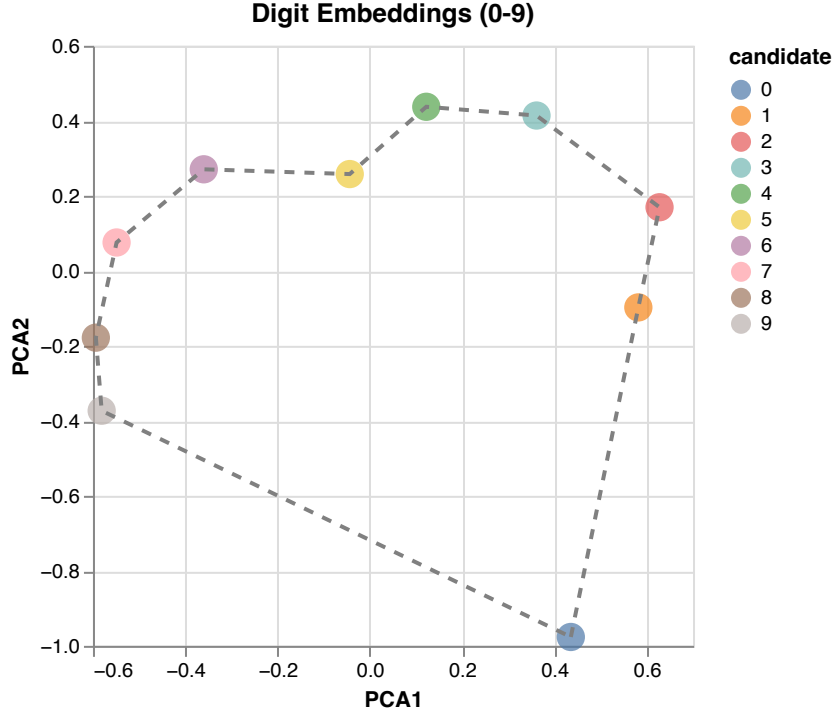


Figure 8: 2D projection of digit unembedding vectors. The embeddings are projected to 2D space using PCA. Each point represents a digit, and the points are connected in numerical order.

Our hypothesis is that transformer networks encode templates for outputs in the unembedding space. The MLP layer then selectively reads these templates based on their linear projection W_{down} . By triggering a specific combination of neurons, each representing a template, the network ensures the logits reach their maximum value for the tokens with the highest probability.

4 Related Work

Our work builds upon and is inspired by several previous works. The logit lens method (nostalgebraist, 2020) is closely related, allowing exploration of the residual stream in transformer networks by treating middle layer hidden states as the final layer output. This provides insights into how transformers iteratively refine their predictions. The author posits that decoder-only transformers operate mainly in a predictive space.

The tuned lens method (Belrose et al., 2023) improves upon the logit lens approach by addressing issues such as biased estimates and not working as well

with some model families. Their key innovation is adding learnable parameters when reading logits from intermediate layers.

The path expansion trick used by (Elhage et al., 2021) decomposes one- and two-layer transformers into sums of different computation paths. Our approach is similar but treats each component independently to avoid combinatorial explosion in larger networks.

(Wang et al., 2022) examines GPT-2 circuits for the Indirect Object Identification task, using the last hidden state norm to determine each layer’s contribution to the output, similar to our residual prism. Their analysis of the difference in logits between candidates is, in fact, very similar to our candidate comparison.

Our findings in the example section align with previous research in several ways. (Mikolov et al., 2013) demonstrate that their Word2vec technique captures relationships between entities, such as countries and their capital cities, as directions in the embedding space. They show that this holds true for various types of relationships. This aligns with our observation of how the gemma-2b model represents country embeddings and capital city unembeddings.

Numerous studies (Mirzadeh et al., 2023; Zhang et al., 2021, 2024) have empirically observed sparse activations in MLP neurons, which is consistent with our MLP analysis. However, the primary focus of these works is on leveraging the sparsity to accelerate model inference rather than interpreting the model’s behavior.

(Geva et al., 2021) suggest that MLP layers in transformer networks act as key-value memories, where the W_{up} matrix (the key) detects input patterns and the W_{down} matrix (the value) boosts the likelihood of tokens expected to come after the input pattern. In our examples, we show that MLP neurons can learn understandable output templates for digit tokens. (Nanda et al., 2023) study how a simple 1-layer transformer network carries out modular addition task. They discover that the network’s MLP neurons use constructive interference of multiple output templates to shape the output distribution, making it peak at the right answer.

5 Conclusion

This paper introduces logit prisms, a simple but effective way to break down transformer outputs, making them easier to interpret. With logit prisms, we can closely examine how the input embeddings, attention heads, and MLP neurons each contribute to the final output. Applying logit prisms to the gemma-2b model reveals valuable insights into how it works internally.

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